

A Macrofluidic Bioreactor to Recreate Hormone Fluctuations *In Vitro*.

Bridget Gomez, Griffin Huneke, Arnav Mankad.

Group #34

Advisors: Dr. Brianne Connizzo, Caitlin Colicchio. Biomedical Engineering, Boston University

Department of Biomedical Engineering
Boston University

April 29, 2025

Abstract

While sex and hormonal differences in tendon mechanobiology are widely recognized, their underlying mechanisms – particularly how cyclical hormonal fluctuations influence tendon homeostasis and extracellular matrix remodeling – remain poorly understood. Current studies utilize a custom-built mechanical loading bioreactor with a simplified discrete model of the murine estrous cycle to test tendon explants under different mechanical and environmental conditions. To better replicate *in vivo* conditions, this project developed a continuous-flow bioreactor system that models hormonal fluctuations *in vitro*. The system employs a MATLAB-controlled peristaltic pump with high-resolution flow control. Three media channels (two hormone stocks and one dilution media) merge into a redesigned reactor well with an integrated passive-mixing system. The mixing system utilizes channel geometry to encourage mixing of the laminar media streams and ensures uniform output into the culture well. Validation of mixing performance and spatial uniformity within the reactor well was achieved through a custom fluorescence-based assay. Fluorescein and Rhodamine-B stocks were used as analogues to hormones, and their relative fluorescent contributions were measured from samples taken throughout the well. This novel bioreactor system overcomes limitations of static or stepwise hormone delivery, offering a platform to study real-time hormonal impacts on tissue mechanics. By closely mimicking *in vivo* conditions, the bioreactor provides a foundation for investigating sex-specific tissue responses.

1. Introduction & Background

While sex and hormonal differences in tendon mechanobiology are widely recognized, their underlying mechanisms – particularly how cyclical hormonal fluctuations influence tendon homeostasis and extracellular matrix remodeling – remain poorly understood. Biological sex has a clear impact on soft tissue homeostasis and recovery in individuals. Women sustain anterior cruciate ligament (ACL) injuries at two to eight times the rate of men in the same sports¹ and menopausal women having overall higher rates of musculoskeletal injury.² These injury rates have been linked to tendon stiffness which is decreased in women compared to men.³ One study supplementing ligaments engineered from human ACL fibroblasts with physiologic concentrations of estrogen showed decreased lysyl oxidase activity, which may be linked to decreased stiffness and increased injury rates.⁴ Research indicates that elevated estrogen levels contribute to this reduced stiffness,⁵ and women have been shown to have cyclic variations in knee laxity that follow the fluctuations in estrogen concentrations throughout the menstrual cycle.⁶ Despite the NIH mandate to consider sex as a biological variable, there is a clear gender bias of samples in existing research with a meta-analysis reporting females represented only 2% of subjects in studies on patellar tendinopathy and only 5% of researchers recognizing the lack of female subjects as a limitation.⁷ This oversight hinders proper understanding of sex differences and their underlying mechanisms.

Further work investigating hormonal cycling is essential for understanding the influence on sex hormones on connective tissues, particularly given the significant effects of hormonal fluctuations on musculotendinous stiffness and tissue behavior.⁸ Estrogen (17β -estradiol) and progesterone (P4) receptors are present in tendon cells,³ suggesting sex hormones have a direct impact on tendon biology and extracellular matrix remodeling. These effects may explain observed associations between hormonal fluctuations and ACL injury. Studies report higher than expected ACL injury rates during the ovulatory phase and lower than expected injury rates during the luteal phase of the menstrual cycle.⁹ The lower risk in the luteal phase could be due to the heightened levels of progesterone in combination with estradiol which, when present together, have been linked to increased collagen synthesis.¹⁰ Additionally, oral contraceptives are reported to reduce injury risk by up to 20%.¹¹ This effect is likely due to the hormonal regulation they provide over the menstrual cycle, minimizing cyclic variations in tendon stiffness and laxity. However, these results are not well corroborated and there is a general lack of consistent and high quality data in this field,¹¹ highlighting the need for study of hormonal cycles.

A custom-built bioreactor system has been developed to study the effects of stress deprivation or mechanical loading on tendon extracellular matrix remodeling in murine tendon explants.^{12,13} This system has been utilized to examine the impacts of hormonal fluctuations on tendon mechanical properties with a discrete hormone cycling protocol. In prior experiments, the estrous cycle was simplified to four hormone concentrations over four days, correlating to each phase of the cycle.¹⁴ The cycle was simulated using supraphysiological hormone levels based on prior murine cell culture experiments.^{15,16} To advance to the next cycle phase, the media was refreshed daily with adjusted hormone levels. The major limitation of this approach is that it fails to capture continuous hormone fluctuations, reducing the cycle to discrete stages rather than reflecting its dynamic nature.

The primary objective of this project was to develop a continuous flow system which allows for dynamic and precise control of hormone concentrations within the bioreactor. A continuous flow bioreactor is able to better replicate physiological fluctuations of the estrous cycle, which allows for targeted study of the hormonal impacts on tendon mechanobiology. One of the primary challenges with small-scale continuous flow is facilitating mixing between parallel streams. There is significant literature describing microfluidic mixing methods and geometries for both active and passive mixing systems, where “active” systems have some external energy input and “passive” systems rely solely on the geometry of the channel and flow characteristics.^{17–19} To reduce the overall complexity of the bioreactor system, we chose to focus primarily on passive mixing systems in our design. The macrofluidic scale of the bioreactor system is a significant departure from the microfluidic scale on which common mixing geometries have been tested, so we sought to validate the mixing behavior on a macroscopic scale and contribute to the base of research on fluidic systems.

This project expands upon the design of the custom-built mechanical loading bioreactor previously described. We have developed a prototype system that utilizes passive mixing geometries and continuously pumps media into and out of the bioreactor. Past work with fluidic devices incorporated fluorescent imaging that could track concentration over space and time.^{20,21} We have used similar techniques to quantify solute levels while developing the mixing and pumping systems to improve our understanding of bioreactor mixing performance. By redesigning the bioreactor to continuously modulate and track media concentration we aimed to provide a more efficient tool to model hormonal conditions for tissue research *in vitro*.

2. Methods

2.1 Dye Preparation and Storage

Powdered forms of Fluorescein²² and Rhodamine B²³ were used to prepare stock solutions. Each dye was dissolved in either deionized water or phosphate buffered saline (PBS) (pH 7.4), depending on the experimental condition. Stock solutions were prepared at appropriate concentrations (e.g., 10 μ M) and stored in light-protected containers covered with aluminum foil. When not in use, all dye solutions were kept refrigerated at 4 °C to prevent photobleaching and degradation.

2.2 Microplate Preparation

Experiments were conducted using black 96-well microplates²⁴, which minimize background fluorescence and cross-talk between wells. Wells containing only solvent (water or buffer) were used as blanks for baseline correction. Figure 1 is an example of a plate set up using a PBS buffer solvent. Each well contains either one dye, a two-dye mixture, plain solvent, or nothing. Row A has the background PBS buffer. Rows B and F have only Rhodamine B dye, serially diluted across the columns. Rows D and H have the same concentration profiles of Fluorescein dye. Rows C and G have 50 μ L of both dyes taken from the adjacent single-dye rows, indicated in yellow to make a row of known mixture concentrations. The single dye wells originally had 150% the desired final volume, so all wells ultimately have 100 μ L of liquid with a minimum of two replicates per condition.

2.3 Fluorescence Measurements

Fluorescence intensity was measured using a BioTek Synergy H1 Multimode Reader microplate reader²⁵. Excitation and emission wavelengths varied since the aim of some experiments was to optimize their values, but results were compared against each other only when reading conditions were the same (settings used are described in 3.1). Instrument settings such as gain, read height, and integration time were kept constant across all runs. Plates were read immediately after preparation to minimize variation due to photobleaching. For the mixing tests, such as the one in Figure 1, the entire plate was read once at the chosen Rhodamine B wavelengths and once with those for Fluorescein.

2.4 Concentration Prediction

From fluorescence measurements, concentrations of solutions were predicted using the dyes' standard curve equations. Serial dilution plates done with single dyes provide a relationship between fluorescent intensity reading and known concentration. This data was linearly fit, and the resulting equations were used to determine concentration of each dye using readings of an unknown solution. A plate was read twice using both dyes' associated excitation and emission wavelengths.

2.5 Data Analysis

Raw fluorescence values were blank-corrected using the average signal from solvent-only wells. Measurements were used to calculate the concentration using the standard curve, then averaged across replicates and analyzed using Microsoft Excel and MATLAB. The percent error was calculated with the averaged value using Equation 1.

$$\%E = \frac{| \text{Measured Concentration} - \text{Known Concentration} |}{\text{Known Concentration}} * 100\% \quad (1)$$

For pump and mixing validation, the measured values were compared with the expected concentration using a one-sample t-test.

2.6 Larger Volume Methodology

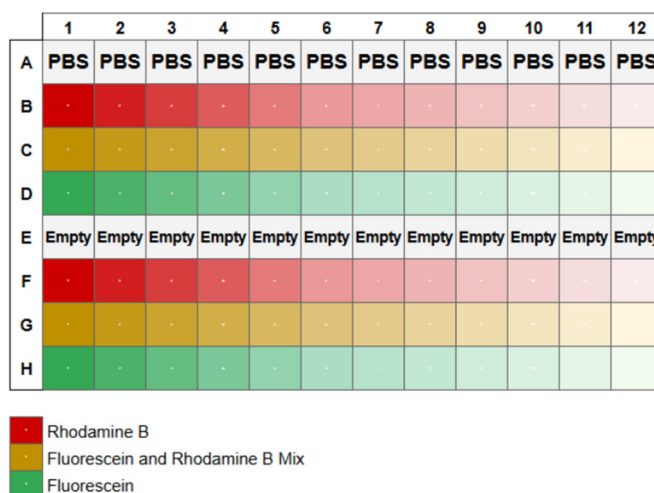


Figure 1. Example plate configuration for testing fluorescent dye mixtures with Rhodamine B (red), Fluorescein (green), and a 1:1 mixture (yellow). Darkness in color represents descending total fluorophore concentration per row.

To evaluate the assay in larger volumes, two-dye solutions were prepared in 50 mL covered containers and 100 μ L aliquots were pipetted into a column of wells for fluorescence measurement. Dye concentrations were determined using previously established standard curves. To apply this method in the well system, single-dye solutions were prepared and pumped into a mixing plate chamber that flows into the open well region. Samples were collected in triplicate from the following locations: (1) the single-dye and buffer stock solutions, and (2) the three outlets of the mixing chamber. Samples taken at the outlets of the mixing chamber were collected while the pump was running in order to assess the efficiency of the mixing chamber. Consistent concentrations across all sampling sites was used as an indication of uniform mixing.

2.7 Well Plate Prototyping

Well plate designs were created using OnShape,²⁶ a cloud-based computer-aided design (CAD) software, as detailed in the results section. Each design consists of two primary components: a mixing chamber and a culture well, connected by a manifold. The mixing chamber was engineered to passively blend three fluids of similar viscosities, representing estrogen, progesterone, and non-hormonal media. Early designs incorporated circular, then triangular, pillars within the chamber to evaluate their effectiveness in disrupting flow and enhancing mixing.²⁷ In subsequent iterations, the pillars were removed, channel widths were reduced to encourage passive fluid convergence, and various geometries, such as serpentine²⁷ and intersecting channels, were tested to evaluate mixing efficiency through fluid collision.

The culture well was designed to collect the mixed solution for downstream testing of hormonal effects on tendon tissue. It included features such as tendon explant grippers and fluid outflow ports. A manifold with one inlet and three outlets connected the mixing chamber to the culture well and was progressively optimized to ensure even distribution of fluid across the entire well.

COMSOL Multiphysics 6.2²⁸ is a software that can model physics-based systems, including fluid flow through a solid interface. The primary objective was to use COMSOL to estimate the output flow rate into the culture well and evaluate how it distributed across the three output channels designed to evenly spread fluid into the well. This allowed assessment of both the mixing uniformity and the effectiveness of manifold design in delivering a balanced flow to all regions of the culture well.

In this study, 2D CAD designs of each mixing chamber were created using OnShape and imported into COMSOL in .stl format for analysis. Simulations were conducted using the Laminar Flow (spf) physics interface to model fluid behavior under low Reynolds number conditions typical of microfluidic environments. A stationary study was used, meaning the software assumed steady-state conditions without accounting for time-dependent changes. The fluid flow rate was standardized at 0.1 mm/s, with minor variations (± 0.05 mm/s) introduced to confirm that fluctuations in velocity did not significantly influence the results. Additionally, COMSOL's *Transport of Diluted Species* module was used to simulate the mixing of three fluids with similar concentrations (0.1 mmol/m³), enabling computational estimation of mixing efficiency within the chamber.

2.8 Well Plate Testing

Benchtop testing of the well design was completed using a Reglo ICC 4-channel digital peristaltic pump,²⁹ with custom MATLAB³⁰ scripts to control the flow protocol. The pump is controlled through a custom MATLAB class which handles the translation of user input into serialized binary strings based on the pump communication protocol specified by the manufacturer. The code is easily adaptable to future experimental implementations and provides fully customizable control of the flow system. The flow system was validated through the following protocols following calibration done to manufacturer guidelines. (1) The pump was used to dispense specified ratios of Fluorescein and Rhodamine B solution into an empty beaker. These mixtures were pipetted in 100 μ L aliquots into a column of wells for microplate analysis described in 2.3. (2) Equal flow rates (1mL/min per channel, net 3mL/min) of Fluorescein, Rhodamine B, and PBS stocks were pumped into the inlet of the well plate, samples were collected from the output of the manifold between the mixing channel and the culture well while the pump was running. The samples were analyzed as described in 2.4-2.5.

3. Results

3.1 Fluorescent Assay Standard Curves

We began by creating standard curves for both Fluorescein and Rhodamine B in water, using excitation/emission wavelengths near their respective peaks (Fluorescein: 460/510 nm; Rhodamine B: 545/567 nm). These standard curves were used to estimate the concentrations of known mixtures prepared in both the

micromolar and picomolar range. However, the calculated concentrations showed very high percent error, over 200% in both ranges, indicating significant inaccuracy in our initial setup (Table 1).

To improve the accuracy of the assay, we made several changes to the experimental design. First, instead of using full serial dilutions across a wide range, we tested plates with a narrower concentration range to reduce issues with plate reader insensitivity at the extremes. Second, we switched the solvent from water to PBS since both dyes are slightly pH sensitive and PBS helps maintain them in a stable pH range, reported to preserve fluorescence intensity. Third, we tested serial dilutions at the pM and μ M scale since the pM concentrations, although being more biologically relevant to hormone levels, are suspected to be too low to register reliably based on the plate reader specifications.

According to a spectral scan, Fluorescein was being weakly excited at Rhodamine B's peak excitation wavelength (545 nm) since Fluorescein excitation slightly overlaps with the Rhodamine B emission window (567 nm). To address this spectral overlap and reduce crosstalk, we shifted the Rhodamine B reading settings to excitation at 551 nm and emission at 580 nm, moving both values higher to reduce interference from Fluorescein. The most recent iteration using the shifted wavelengths and a PBS solvent gave the most accurate results so far (Table 1). The standard curves associated with these calculations are shown in Figure 3. The standard curves used for the fourth row of testing in Table 1 were linearly fit to $y = 43603x + 16584$ (Fluorescein) and $y = 29245x + 7793$ (Rhodamine B). Two outliers from Rhodamine B were taken out at higher concentrations and both linear fits were successful with $R^2 = 0.999$. These curves were used to test all fluorescent samples for mixing and pump validation tests.

Table 1. Summary of the averaged accuracy rates of duplicate different microplate experimental procedures at predicting concentration

Experimental setup	Average Prediction %E
Water, pM range	Fluorescein: 1932%, SD = ± 1326.36 pM Rhodamine B: 137844%, SD = ± 386951.64 pM
Water, μ M range	Fluorescein: 46.62%, SD = ± 18.69 μ M Rhodamine B: -432.23%, SD = ± 17.87 μ M
PBS, pM range	Fluorescein: 26.14%, SD = ± 12.98 pM Rhodamine B: 28537%, SD = ± 22.98 pM
PBS, μ M range, non-peak wavelengths	Fluorescein: 24.93%, SD = ± 0.4 μ M Rhodamine B: 22.83%, SD = ± 0.48 μ M

While Fluorescein and Rhodamine B were used for well and pump testing, other infrared fluorophores (SNAP-JFX650, Alexa Fluor 647) had preliminary results indicating they could be viable replacements for Rhodamine B with further testing, with higher accuracy rates in low concentration mixtures. Their spectra are further apart from Fluorescein, allowing for higher gain settings and measurement at peak intensity wavelengths, rather than the required offset to avoid crosstalk.

3.2 Pump Validation

The pump was programmed to create solutions with the following ratios of each dye from stocks: 1:1, 1:4, 4:1, 1:10, and 10:1. The dispensed solution was not different from the expected concentration for all ratios except the last, which was statistically different.

3.3 Mixing Plate Simulation

In total, 16 unique well plate designs were modeled in OnShape, with flow rates and mixing efficiency evaluated through both software-based simulations and benchtop experiments. Figure 2a shows the sixth iteration of the well plate, featuring three inlets and circular posts to enhance passive mixing by disrupting flow. Fluids were designed to converge into a single output channel before entering a three-way manifold, which aimed to evenly distribute flow into the culture well. To help balance flow rates across all three outlets, the channel geometry was adjusted so that the straightest path had the longest length, compensating for the natural preference of fluids to follow the path of least resistance.

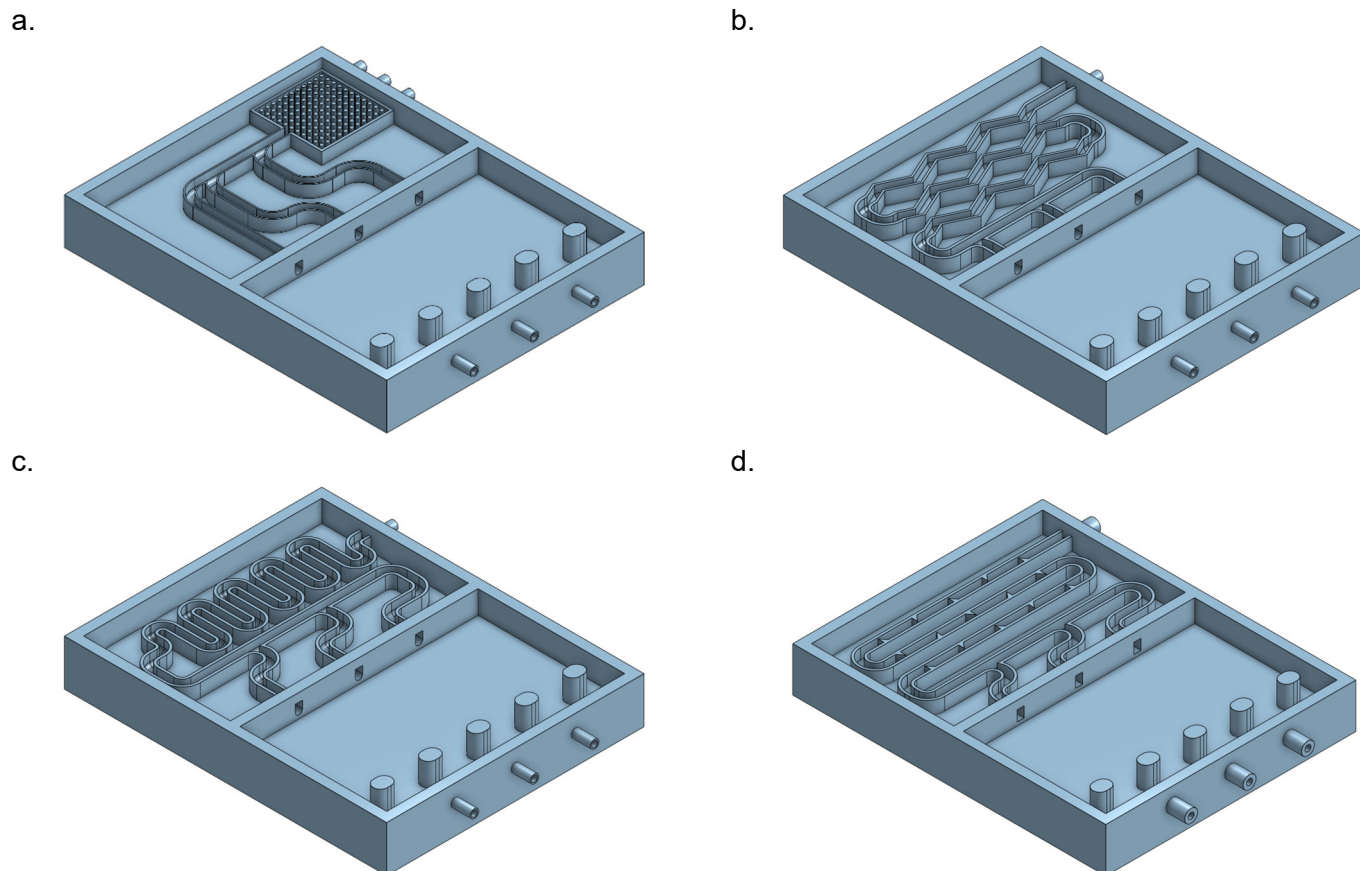


Figure 2. CAD models of four different bioreactor well plate designs. (a) shows the earliest iteration, followed by (b), (c), and (d) representing progressively refined designs. The overall dimensions of each well plate, determined by bioreactor specifications, are $134 \times 107 \times 19.05$ mm. The mixing chambers and manifolds are 4.5 mm deep, while the culture wells are 9.525 mm deep.

Figure 2b depicts the tenth iteration, which prioritizes mixing over distribution using a large serpentine channel with straight-edged walls to encourage fluid interaction. Figure 2c, the thirteenth iteration, combines the serpentine geometry from Figure 2b with the effective manifold design from Figure 2a, aiming to optimize both mixing and even fluid distribution. Finally, Figure 2d, the sixteenth iteration, introduces sharp triangular posts within the mixing chamber to promote mixing through secondary flow due to the sharp corner.³¹ Narrow gaps between the posts and the channel walls force fluids through constricted regions, further enhancing mixing. Additionally, the three iterations utilized cross-junctions outside of the well for preliminary mixing. The manifold geometry was retained due to its demonstrated effectiveness in prior iterations.

From computational mixing analysis shown in Figure 3, it is evident that only the earliest iteration (Figure 3a) fails to effectively mix the three fluids before entering the culture well. In contrast, the other three designs demonstrate uniform distribution throughout the mixing chamber under steady-state conditions.

From computational flow speed analysis shown in Figure 4, it is clear that the manifold designs in Figures 6b and 6d achieve an equal distribution of mixed fluid across the culture well, as indicated by uniform velocities across the three output channels. In contrast, Figure 4a shows a higher fluid velocity in the far-left channel compared to the others, while Figure 4c exhibits the highest velocity in the central channel.

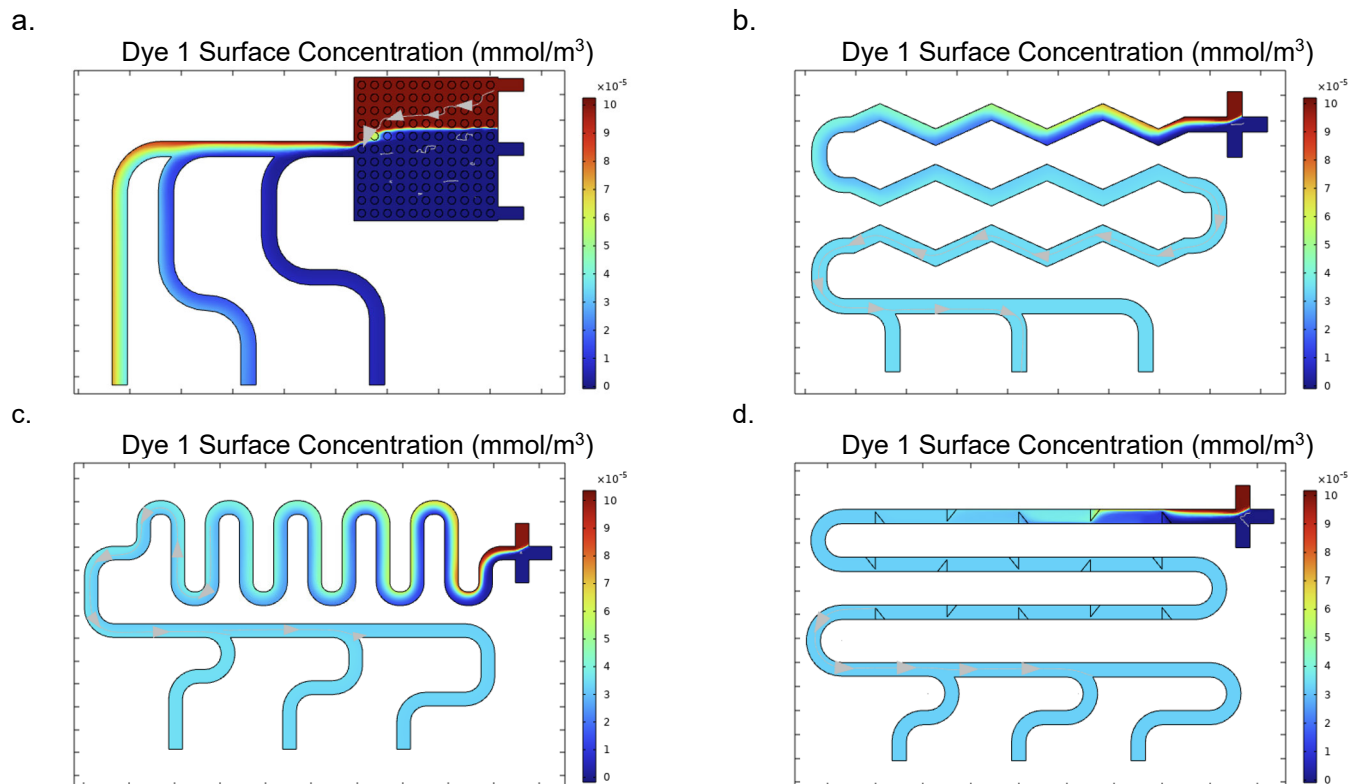


Figure 3. Results from COMSOL mixing simulations which show the distribution of one fluid throughout the mixing chamber in steady-state conditions. (a), (b), (c), and (d) correspond to progressively improved mixing chamber designs that enhance mixing efficiency through geometric modifications.

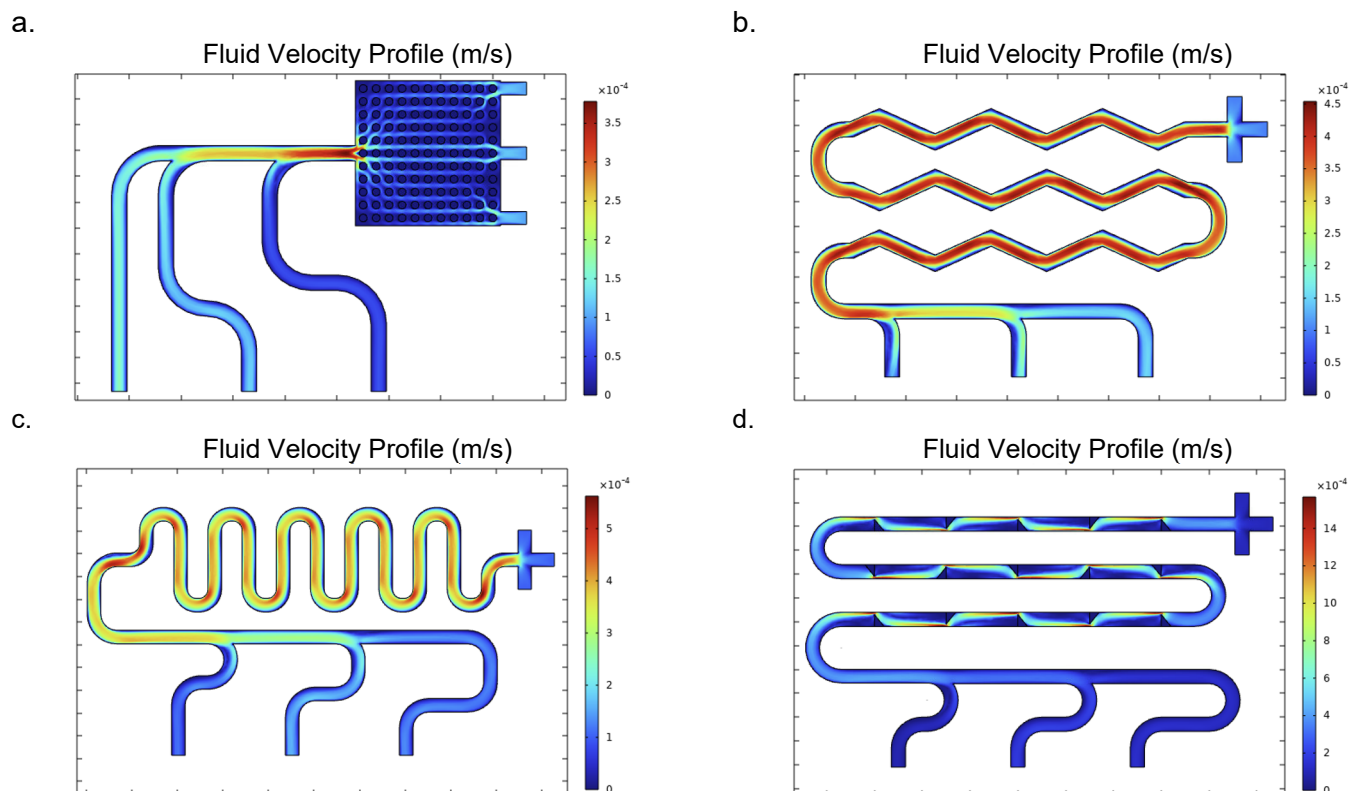


Figure 4. Results from COMSOL flow simulations showing the fluid velocity distribution across the three outlet channels of the manifold. (a), (b), (c), and (d) correspond to different mixing chamber and manifold configuration iterations from figure 2.

3.4 Mixing Plate Validation

Benchtop results (Figure 5) show a significant difference in qualitative mixing performance compared to COMSOL simulations. Flow streams stayed separated for much longer than predicted and did not fully mix from a visual perspective in any of the well iterations tested. Our final design shown in Figure 5d visually appears to have the most impact on the mixing of the input streams. Designs used in Figures 7a and 7c had little to no qualitative impact on the streams, with both primary color channels remaining distinctly separate throughout the mixing stage. Figure 5b showed more promising results visually, however the outlet still has clear distinctions between color channels.

Fluorescent samples from mixing plate (a) were not collected as the design and initial testing was completed before our fluorescent methodology was complete. Visual colorimetric analysis showed a clear lack of mixing, so the design was iterated upon before quantitative testing was done. Sampling at the outlets of the fluorescent solutions mixing plates (b) and (c) showed significant differences from an expected equal contribution at each channel, indicating that the mixing plates were inefficient ($p\text{-value} < 0.05$). Since the measurements at the inlet were not significantly different from expected values, the mixing system and split into the manifold outlet appears to be the component preventing proper mixing. However, mixing plate (d) achieved proper mixing in two of its outlets, with their samples not significantly different from expected.

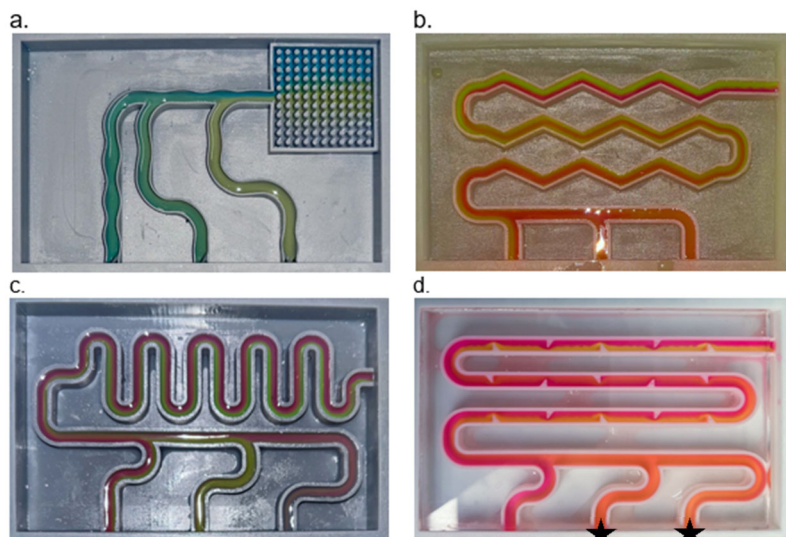


Figure 5. Benchtop flow evaluation using colorimetric (a) and fluorescent (b, c, d) dyes to evaluate mixing performance qualitatively and quantitatively. * indicates fluorescent values were not significantly different from values due to expected equal contributions

4. Discussion

We have successfully developed a continuous flow system and accompanying fluorescence assay for use in flow validation. Despite developing sixteen different mixing system iterations, we were unable to fully meet our initial goal of producing a uniformly mixed flow into the culture well of the bioreactor. Due to the relatively large scale of flow compared to typical microfluidic systems, passive mixing geometries which have been well validated for microfluidic scales were largely ineffective for our application. This is likely due to the channel size and volume of flow, which are roughly an order of magnitude larger than mixing geometries have been tested on. Despite the larger volume and flow rate, our flow system maintained laminar behavior which prevented the passive diffusion mixing of each input stream. The faster flow rates likely played a role in the discrepancy between COMSOL simulated results and observed physical behavior. At a net flow rate of 3 mL/min across all channels, our benchtop tests were producing a surface velocity of ~ 24 cm/s which is much faster than the 0.1 mm/s that COMSOL was capable of simulating. Slowing benchtop flow to 0.3 mL/min (~ 2.4 cm/s) did not produce visual improvements in flow separation or improve comparisons to COMSOL mixing simulations.

Our final mixing design showed promising results, with two of three outputs producing solutions which were not significantly different from expected results, indicating even mixing at these outlets. The performance of the design may improve if the length of the mixing channel is increased, and the size and arrangement of the sharp corners are adjusted for greatest mixing impact. The overall size of our system was constrained to fit into the dimensions of the current bioreactor system to reduce experimental changes needed to utilize the continuous flow system. We decided to focus on a passive mixing system to minimize the overall complexity of the bioreactor system. However, given the difficulties of producing an effective passive system, it may be beneficial to transition to an active mixing system despite the added complexity to ensure fully mixed media. An active system would add operational and setup complexity, as well as introduce another sterility concern for tissue culture. If a passive mixing system is required, it may be beneficial to increase the available area for mixing or separate the mixing process from the main culture well and onto a separate plate. This would allow for elongation of the mixing

channel which will allow more interaction between the fluid and mixing geometry and allow for more diffusion mixing between the different streams.

Regardless of our challenges with designing an effective mixing system, the pump system has shown promising results for future use and could provide an effective platform for accurate hormonal dosing over long periods of time once an acceptable mixing solution is developed. Additionally, our design progress on the mixing system will prove a useful starting point for future iterations to improve mixing performance. This prototype will provide a valuable contribution towards recreating physiologic hormone fluctuations *in vitro* for studying hormonal impacts on tendon biology. Continuation of this project will contribute to furthering areas of women's health research which are often underrepresented.

5. References

1. Wojtys, E. M., Huston, L. J., Boynton, M. D., Spindler, K. P. & Lindenfeld, T. N. The Effect of the Menstrual Cycle on Anterior Cruciate Ligament Injuries in Women as Determined by Hormone Levels. *Am. J. Sports Med.* **30**, 182–188 (2002).
2. Enns, D. L. & Tiidus, P. M. The Influence of Estrogen on Skeletal Muscle. *Sports Med.* **40**, 41–58 (2010).
3. Hansen, M. & Kjaer, M. Sex Hormones and Tendon. in *Metabolic Influences on Risk for Tendon Disorders* (eds. Ackermann, P. W. & Hart, D. A.) 139–149 (Springer International Publishing, Cham, 2016). doi:10.1007/978-3-319-33943-6_13.
4. Lee, C. A. *et al.* Estrogen inhibits lysyl oxidase and decreases mechanical function in engineered ligaments. *J. Appl. Physiol.* **118**, 1250–1257 (2015).
5. Chidi-Ogbolu, N. & Baar, K. Effect of Estrogen on Musculoskeletal Performance and Injury Risk. *Front. Physiol.* **9**, 1834 (2019).
6. Shultz, S. J., Gansneder, B. M., Sander, T. C., Kirk, S. E. & Perrin, D. H. Absolute serum hormone levels predict the magnitude of change in anterior knee laxity across the menstrual cycle. *J. Orthop. Res. Off. Publ. Orthop. Res. Soc.* **24**, 124–131 (2006).
7. Mondini Trissino da Lodi, C., Landini, M. P., Asunis, E. & Filardo, G. Women Have Tendons... and Tendinopathy: Gender Bias is a 'Gender Void' in Sports Medicine with a Lack of Women Data on Patellar Tendinopathy-A Systematic Review. *Sports Med. - Open* **8**, 74 (2022).
8. Eiling, E., Bryant, A. L., Petersen, W., Murphy, A. & Hohmann, E. Effects of menstrual-cycle hormone fluctuations on musculotendinous stiffness and knee joint laxity. *Knee Surg. Sports Traumatol. Arthrosc.* **15**, 126–132 (2007).
9. Wojtys, E. M., Huston, L. J., Boynton, M. D., Spindler, K. P. & Lindenfeld, T. N. The Effect of the Menstrual Cycle on Anterior Cruciate Ligament Injuries in Women as Determined by Hormone Levels. *Am. J. Sports Med.* **30**, 182–188 (2002).
10. Hansen, M. Female hormones: do they influence muscle and tendon protein metabolism? *Proc. Nutr. Soc.* **77**, 32–41 (2018).
11. Herzberg, S. D. *et al.* The Effect of Menstrual Cycle and Contraceptives on ACL Injuries and Laxity: A Systematic Review and Meta-analysis. *Orthop. J. Sports Med.* **5**, 2325967117718781 (2017).
12. Aggouras, A. N., Stowe, E. J., Mlawer, S. J. & Connizzo, B. K. Aged Tendons Exhibit Altered Mechanisms of Strain-Dependent Extracellular Matrix Remodeling. *J. Biomech. Eng.* **146**, 071009 (2024).
13. Connizzo, B. K., Piet, J. M., Shefelbine, S. J. & Grodzinsky, A. J. Age-associated changes in the response of tendon explants to stress deprivation is sex-dependent. *Connect. Tissue Res.* **61**, 48–62 (2020).
14. Sander, A. M. Uncovering the individual impacts of sex, hormones, and hormone cycling on tendon remodeling. (Boston University, 2024).
15. Lee, C.-Y. *et al.* The combined regulation of estrogen and cyclic tension on fibroblast biosynthesis derived from anterior cruciate ligament. *Matrix Biol.* **23**, 323–329 (2004).
16. Tashjian, R. Z. *et al.* The effect of estrogen-like compound on rotator cuff tendon healing in a murine model. *J. Orthop. Res.* **39**, 2711–2724 (2021).
17. Li, Z. *et al.* A review of microfluidic-based mixing methods. *Sens. Actuators Phys.* **344**, 113757 (2022).
18. Wang, X., Liu, Z., Cai, Y., Wang, B. & Luo, X. A cost-effective serpentine micromixer utilizing ellipse curve. *Anal. Chim. Acta* **1155**, 338355 (2021).
19. Chen, K., Lu, H., Sun, M., Zhu, L. & Cui, Y. Mixing enhancement of a novel C-SAR microfluidic mixer. *Chem. Eng. Res. Des.* **132**, 338–345 (2018).
20. Aryal, P., Hefner, C., Martinez, B. & S. Henry, C. Microfluidics in environmental analysis: advancements, challenges, and future prospects for rapid and efficient monitoring. *Lab. Chip* **24**, 1175–1206 (2024).
21. Bató, L. & Fürjes, P. Diffusion coefficient measurement with fluorescent detection in free-diffusion based microfluidics. *Microfluid. Nanofluidics* **28**, 56 (2024).
22. Fluorescein Sodium Salt Fluorescent Tracer NaFluo.
<https://www.sigmaaldrich.com/US/en/product/sigma/46960>.
23. Rhodamine B = 95 HPLC 81-88-9. <https://www.sigmaaldrich.com/US/en/product/sigma/r6626>.
24. 96 Well Black Plate, Non-Treated Surface, No Lid, Non-Sterile, Pack of 10.
<https://www.thermofisher.com/order/catalog/product/237108>.
25. Multimode Microplate Reader, BioTek Synergy H1 | Agilent.
<https://www.agilent.com/en/product/microplate-instrumentation/microplate-readers/multimode-microplate-readers/biotek-synergy-h1-multimode-reader-1623193>.

26. Onshape. PTC (2015).
27. Kheirkhah Barzoki, A. Optimization of passive micromixers: effects of pillar configuration and gaps on mixing efficiency. *Sci. Rep.* **14**, 16245 (2024).
28. COMSOL Multiphysics. COMSOL Inc.
29. Masterflex® Ismatec® Reglo ICC Digital Pump, 4-Channel, 12-Roller; 100 to 240 VAC, 50/60 Hz. <https://www.avantorsciences.com/us/en/product/NA5139093/masterflex-ismatec-reglo-independent-channel-control-icc-peristaltic-pumps-avantor>.
30. MATLAB. The MathWorks, Inc. (2023).
31. Fan, L.-L., Zhu, X.-L., Zhao, H., Zhe, J. & Zhao, L. Rapid microfluidic mixer utilizing sharp corner structures. *Microfluid. Nanofluidics* **21**, 36 (2017).